

# Sums, Quotients, and Complexes

Math 5250 — Week 2

# Direct Sums

Let  $V, W$  be vector spaces over a field  $\mathbf{F}$ . The **direct sum**

$$V \oplus W = \{(v, w) : v \in V, w \in W\}$$

is a vector space with  $(v, w) + (v', w') = (v + v', w + w')$  and  $a(v, w) = (av, aw)$ .

- ▶ The direct sum of a finite family  $\{V_i\}_{i=1}^m$  is the set of sequences  $(v_1, \dots, v_m)$ ,  $v_i \in V_i$ .
- ▶ For an index set  $I$ ,  $\bigoplus_{i \in I} V_i$  is the set of  $f : I \rightarrow \bigcup_i V_i$  with  $f(i) \in V_i$ , nonzero for only finitely many  $i$ .
- ▶  $\mathbf{F}$  is a one-dimensional vector space over itself, and  $\mathbf{F}^n = \bigoplus_{i=1}^n \mathbf{F}$ .

# Direct Sums: Dimension

$V$  and  $W$  sit inside  $V \oplus W$  as  $(v, 0)$  and  $(0, w)$ , intersecting only in  $(0, 0)$ .

**Proposition.** If  $V, W$  are finite dimensional of dimensions  $n, m$ , then  $V \oplus W$  is finite dimensional of dimension  $n + m$ ; if  $v_1, \dots, v_n$  is a basis of  $V$  and  $w_1, \dots, w_m$  a basis of  $W$ , then

$$(v_1, 0), \dots, (v_n, 0), (0, w_1), \dots, (0, w_m)$$

is a basis of  $V \oplus W$ .

6  $v = \sum a_i v_i \quad w = \sum b_j w_j$  and  $a_i = 0$   
 $b_j = 0$

$$(v, w) = \sum a_i (v_i, 0) + \sum b_j (0, w_j) = \sum a_i (v_i, 0) + \sum b_j (0, w_j) = (\sum a_i v_i, \sum b_j w_j) = (v, w)$$

# Subspace Sums

Let  $U, W$  be subspaces of  $V$ . The **sum**

$$U + W = \{u + w : u \in U, w \in W\}$$

is a subspace of  $V$ .

$\text{Span}(u_1, \dots, u_n)$   
 = smallest subspace  
 containing  $u_1, \dots, u_n$   
 $\langle u_i \rangle = 1$  dim'd subspace  
 spanned by  $u_i$ .

$$\langle u_1 \rangle + \langle u_2 \rangle + \dots + \langle u_n \rangle$$

- ▶ It is the smallest subspace of  $V$  containing both  $U$  and  $W$ .
- ▶ If  $u_1, \dots, u_n$  span  $U$  and  $w_1, \dots, w_k$  span  $W$ , then
  - ▶  $u_1, \dots, u_n, w_1, \dots, w_k$  span  $U + W$ .
- ▶  $\dim(U + W) \leq \dim(U) + \dim(W)$ , and this can be strict (e.g.  $U = W$ ).

$\mathbb{R}^4$

$U$

$$\begin{pmatrix} 1, 0, 0, 0 \\ 0, 1, 0, 0 \end{pmatrix}$$

$W$

$$\begin{pmatrix} 0, 0, 1, 0 \\ 0, 0, 0, 1 \end{pmatrix}$$

# Algorithm to Compute a Basis for $U + W$

$$U, W \subseteq V \quad E = \{v_1, \dots, v_n\} \text{ basis for } V \quad r+s = \dim(U) + \dim(W) \\
u_1, \dots, u_r \text{ basis for } U \quad [u_i]_E \quad n \uparrow \downarrow \quad \underbrace{r}_{\dim(U)} \quad \underbrace{s}_{\dim(W)} \quad \underbrace{n}_{\dim(U+W)} \\
w_1, \dots, w_s \quad [w_j]_E \quad \left[ \begin{array}{c|c} [u_i]_E & [w_j]_E \end{array} \right]$$

1. Choose a basis for  $V$  and spanning sets for  $U$  and  $W$  as above. Form the matrix  $A$  whose columns are the  $u_i$  and  $w_j$  in the chosen basis.
2. Row-reduce  $A$  to  $A'$ .
3. The columns of  $A$  corresponding to pivot columns of  $A'$  form a basis for  $U + W$ .
4. This basis is a subset of the union of the  $u_i$  and  $w_j$ .

$$\dim(U+W) = rk \left[ \begin{array}{c|c} [u_i]_E & [w_j]_E \end{array} \right]$$

## Internal Direct Sums

Suppose  $x \in U+W$  is uniquely  $u+w$   $u \in U, w \in W$

$$f: U+W \rightarrow U \oplus W$$

$$f(x) = (u, w)$$

$$U \oplus W \rightarrow U+W$$

$$(u, w) \mapsto u+w$$

**Definition.** If every  $x \in U + W$  has a *unique* representation  $x = u + w$  with  $u \in U, w \in W$ , then  $U + W$  is an (internal) direct sum, isomorphic to  $U \oplus W$ .

**Proposition.**  $U + W$  is a direct sum if and only if either:

a.  $U \cap W = 0$ , or

→ b.  $\dim(U) + \dim(W) = \dim(U + W)$ .

$$\dim(U \oplus W) = \dim(U) + \dim(W)$$

$$\dim(U+W)$$

$$\boxed{\dim(U) + \dim(W) = \dim(U+W) + \dim(U \cap W)}$$

$$U, W \subseteq V$$

$$U \oplus W \quad U+W \subseteq V$$

$$f: (u, w) \longrightarrow u+w$$

$$f((u, w) + (u', w')) = f((u+u', w+w')) \\ \approx u+u' + w+w'$$

$$\text{null}(f) = \{ (u, w) \mid f(u, w) = 0 \} = \{ (u, w) \mid u+w=0 \} \\ u = -w \quad u \in W \cap U$$

$$u \in U \cap W \\ (u, -u) \in \text{null}(f) \\ U \cap W \subseteq \text{null}(f)$$

$$\text{null}(f) = U \cap W \\ \text{image}(f) = U+W$$

$$\underline{U \cap W} \quad U \oplus W \longrightarrow U+W$$

$$\dim(U \cap W) = \text{nullity of } f$$

$$\dim(U+W) = \text{rk}(f)$$

$$\dim(U \oplus W) = \dim(U+W) + \dim(U \cap W)$$

# A Dimension Formula

**Proposition.** For subspaces  $U, W$  of  $V$ ,

$$\dim(U+W) = \dim(U \oplus W) - \dim(U \cap W) = \cancel{\dim(U) + \dim(W)} - \dim(U \cap W) \\ \approx \dim(U) + \dim(W) - \dim(U \cap W)$$

# Equivalence Relations

**Definition.** An *equivalence relation* on  $X$  is a relation  $x \sim y$  satisfying:

1.  $x \sim x$  for all  $x$ .
2.  $x \sim y \implies y \sim x$ .
3.  $x \sim y, y \sim z \implies x \sim z$ .

An equivalence relation partitions  $X$  into equivalence classes  $[x] = \{y \in X : y \sim x\}$ ; distinct classes are disjoint.

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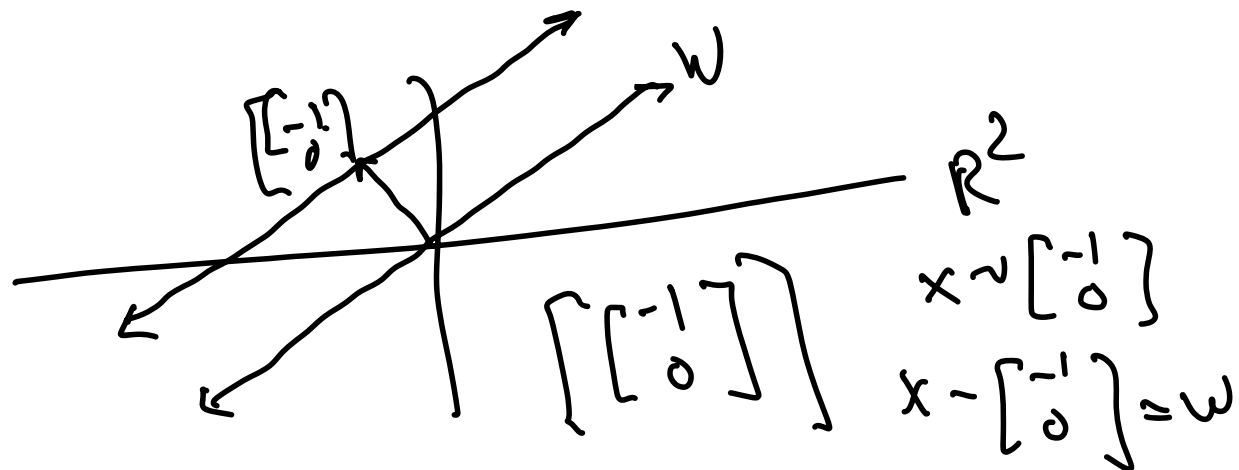
Fix  $N > 0$   $a \equiv b \pmod{N} \iff N \mid a - b$

$[a]$  the set of these  $a' = qN + r$

$[0], [1], \dots, [N-1]$   $0 \leq r < N$

$N \mid a - r$

# Quotient Spaces



Let  $V$  be a vector space over  $F$  and  $W \subset V$  a subspace. Define

$$x \sim y \iff x - y \in W.$$

$$x = \begin{bmatrix} -1 \\ 0 \end{bmatrix} + w$$

**Definition.** The set of equivalence classes  $[x]$  is the **quotient space**  $V/W$ , a vector space over  $F$  with

$$\rightarrow [x] + [y] = [x + y], \quad a[x] = [ax].$$

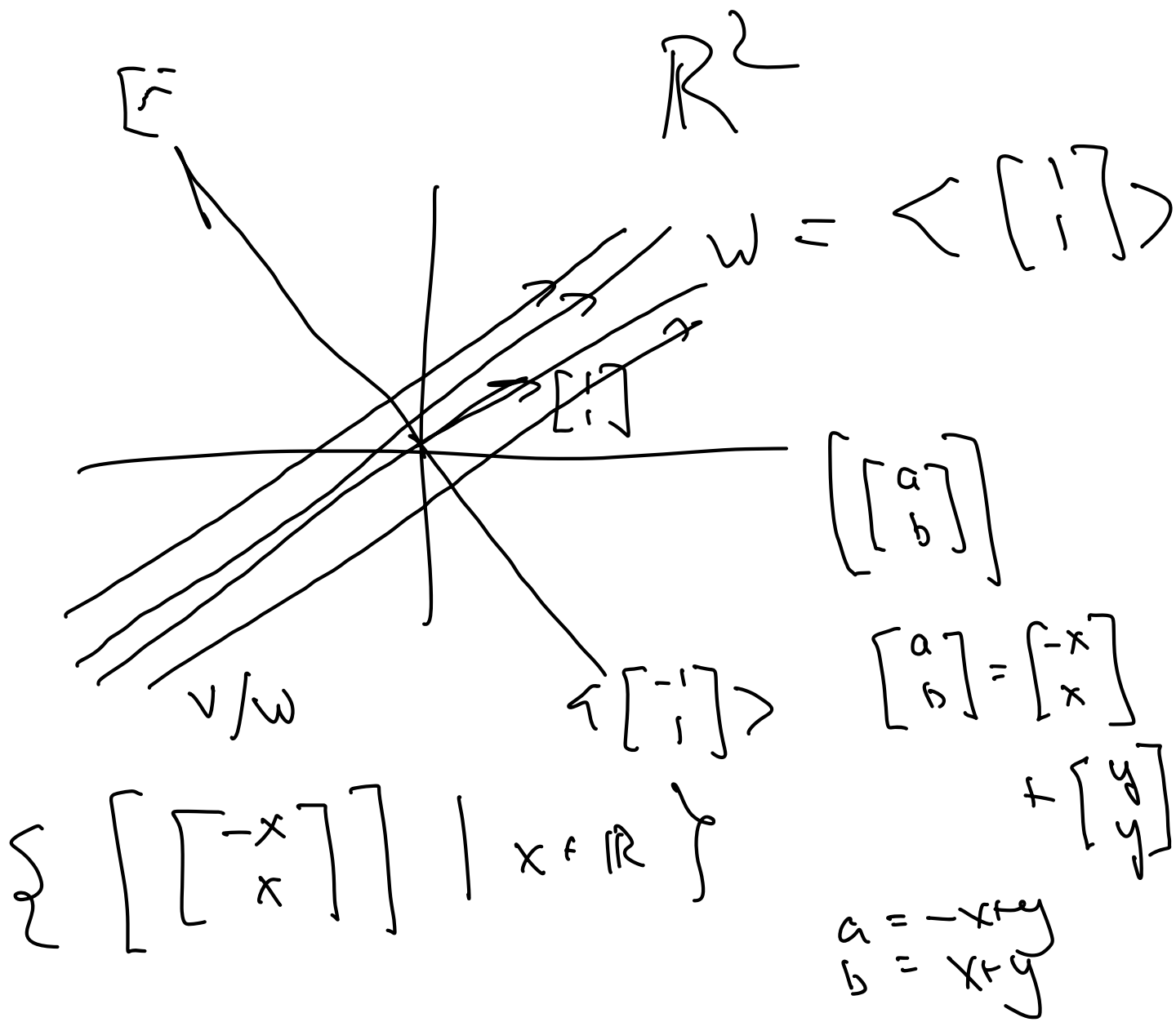
$$[x'] = [x] \quad [y'] = [y]$$

$$x' = x + w_1$$

$$y' = y + w_2$$

$$x' + y' = x + y + \underbrace{(w_1 + w_2)}_{\in W}$$

$$[x' + y'] = [x + y]$$




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$v/w$  is 1-dim'l

$\begin{bmatrix} a \\ b \end{bmatrix}$  is a basis as long as  $a \neq b$ .

# Quotient Spaces: Dimension

**Proposition.** If  $V$  is finite dimensional and  $W \subset V$ , then

$$\dim(V/W) = \dim(V) - \dim(W).$$

(Via the linear map  $f : V \rightarrow V/W$ ,  $f(v) = [v]$ , which is onto with  $\text{null}(f) = W$ .)

$$f(v_1 + v_2) = [v_1 + v_2] = [v_1] + [v_2] = f(v_1) + f(v_2)$$

$f$  surjective because every elt of  $V/W$  is  $[v]$

$$\text{null}(f) = W \quad f(v) = [v] = [0] \iff \underset{v \in W}{v} = g \in W$$

# Universal Property of Quotients

**Proposition.** Let  $f : V \rightarrow W$  be linear and  $H \subset V$  contained in  $\ker(f)$ . Then there is a unique linear map  $\bar{f} : V/H \rightarrow W$  with  $\bar{f}([v]) = f(v)$  for all  $v \in V$ .

The subspaces of  $V/H$  correspond exactly to the subspaces of  $V$  containing  $H$ .

$$f : V \rightarrow W$$

$$H = \ker(f)$$

$$\bar{f} : V/H \longrightarrow W$$

$$\bar{f}([v]) = f(v)$$

# Complexes

**Definition.** A *complex* of vector spaces is a sequence  $W^i$  with linear maps

$$\dots \rightarrow W^i \xrightarrow{d_i} W^{i+1} \xrightarrow{d_{i+1}} W^{i+2} \rightarrow \dots$$

satisfying  $d_{i+1}d_i = 0$  for all  $i$ . The complex is *bounded* if all but finitely many  $W^i$  are zero.

Let  $Z^i = \ker(d_i)$  (the **cocycles**) and  $B^i = d_{i-1}(W^{i-1})$  (the **coboundaries**). The  $i^{th}$  **cohomology** is  $H^i(W^\cdot) = Z^i/B^i$ .

# Euler Characteristic

For a bounded complex of finite dimensional spaces, the **Euler characteristic** is

$$\chi(W^\bullet) = \sum_i (-1)^i \dim(W^i).$$

**Proposition.**

$$\chi(W^\bullet) = \sum_i (-1)^i \dim H^i(W^\bullet).$$

# Exact Complexes

**Definition.** A complex is **exact** if all its cohomology groups are zero; equivalently  $\ker(d^i) = \text{image}(d^{i-1})$  for all  $i$ .

A **short exact sequence** is a 5-term exact complex

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0,$$

so  $A \rightarrow B$  is injective,  $B \rightarrow C$  is surjective, and  $C \cong A/B$ .

# Homology

For a complex indexed the other way,  $d_i : W_i \rightarrow W_{i-1}$ :

- ▶  $Z_i = \ker(d_i)$  is the space of  $i$ -**cycles**.
- ▶  $B_i = \text{image}(d_{i+1})$  is the space of  $i$ -**boundaries**.
- ▶  $H_i(W^\cdot) = Z_i/B_i$  is the  $i^{th}$  **homology group**.

# Maps of Complexes

**Definition.** A map of complexes  $f : \{W^\bullet\} \rightarrow \{V^\bullet\}$  is a family  $f_i : W^i \rightarrow V^i$  with  $f_{i+1} \circ d_i = d_i \circ f_i$ .

Such an  $f$  carries cocycles to cocycles and coboundaries to coboundaries, and therefore induces the **induced map on cohomology**

$$\bar{f} : H^i(W^\bullet) \rightarrow H^i(V^\bullet).$$